The Admissible Observable Class: Gauge-Free Transport Uniqueness of W_{obs} on Real H1 Data

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Abstract

The main vulnerability of the real-data RG pipeline is the apparent freedom in choosing W_{obs}. This paper replaces a coordinate-level definition of W_{obs} by an admissible-class definition. Using the H1 O4a 16 kHz strain segment, four observable representations are compared: primitive spectral bands, dyadic spectral bands, spectral moments, and autocorrelation coordinates. The primitive and dyadic spectral bases both pass the admissibility gate: no 10^{-3} Fisher hole, transport median below 10^{-8}, and relative logarithmic placement error below $2\cdot10^{-3}$. Spectral moments reproduce the placement proxy well but fail transport coherence, while autocorrelation fails placement and null criteria. Thus W_{obs} is not a unique coordinate formula. The supported object is the equivalence class $[W_{\text{obs}}]_{\text{adm}}$ of gauge-free, transport-stable, placement-preserving spectral observable representations.

1 Problem

The earlier real-data paper showed that the primitive spectral observable basis is rank 9 at threshold 10^{-3} and has no primitive Fisher hole. A reviewer can still ask why this W_{obs} was selected. The correct answer is not uniqueness of coordinates, but uniqueness of an admissible transport geometry.

2 Admissible Observable Class

Definition 2.1 (Admissibility gate)**.** *A window observable representation W is admissible for the present placement bridge if it satisfies:*

(i) *no Fisher hole at threshold 10^{-3} in its primitive basis;*

(ii) *median resolved-sector transport distortion below 10^{-8} across internal $256\,\mathrm{s}$ partitions;*

(iii) *relative logarithmic reconstruction/prediction error for ρ_{obs} below $2\cdot10^{-3}$;*

(iv) *spectral accessibility: the representation contains enough spectral information to define or recover the quadratic placement functional.*

Definition 2.2 (Admissible observable class)**.** *The object $[W_{\text{obs}}]_{\text{adm}}$ is the class of all observable representations satisfying the admissibility gate and equivalent under admissible reparameterizations preserving the resolved transport and placement geometry.*

3 Theorem Candidate

Theorem 3.1 (Gauge-null rejection)**.** *If an apparent Fisher null disappears after removing duplicate or algebraically-derived coordinates, then it is a coordinate-gauge null and not a primitive physical blind sector.*

Proof. The augmented observable image lies on the graph of a deterministic map over primitive coordinates. Normal directions to this graph are coordinate constraints. If the null disappears in the primitive basis, it cannot be a primitive unresolved direction. □

Proposition 3.1 (Admissible spectral representations on H1)**.** *On the tested H1 segment, both primitive spectral bands and dyadic spectral bands satisfy the admissibility gate. Spectral moments and autocorrelation do not.*

Proof. The measured metrics are listed in Table 1. Primitive and dyadic bases have zero 10^{-3} holes, transport medians $8.579 \cdot 10^{-16}$ and $7.493 \cdot 10^{-16}$, and placement errors $1.580 \cdot 10^{-3}$ and $1.556 \cdot 10^{-3}$. Spectral moments have two 10^{-3} holes and transport median $6.848 \cdot 10^{-1}$. Autocorrelation has three holes and placement error $3.590 \cdot 10^{-3}$. □

Theorem 3.2 (Admissible transport representation theorem, computational form)**.** *For this H1 segment and this placement functional, the admissible observable object is not a coordinate vector W_{obs} but the class $[W_{\mathrm{obs}}]_{\mathrm{adm}}$. Elements of $[W_{\mathrm{obs}}]_{\mathrm{adm}}$ determine a transport-stable placement geometry; non-admissible replacements may reproduce some scalar placement statistics but do not preserve the complete transport geometry.*

Computational proof. The primitive and dyadic spectral representations both pass the admissibility gate. Spectral moments have excellent placement R^2 but fail transport by many orders of magnitude. Autocorrelation has weaker placement recovery and nonzero holes. Therefore placement prediction alone is insufficient. The admissible object is selected by the joint gate of gauge-free rank, transport coherence, and placement preservation. □

4 Numerical Admissibility Test

Basis	Rank 10^{-3}	Holes	Transport median	ρ rel. error	Pass
Primitive bands	9	0	$8.579 \cdot 10^{-16}$	$1.580 \cdot 10^{-3}$	yes
Dyadic bands	9	0	$7.493 \cdot 10^{-16}$	$1.556 \cdot 10^{-3}$	yes
Spectral moments	7	2	$6.848 \cdot 10^{-1}$	$6.909 \cdot 10^{-4}$	no
Autocorrelation	6	3	$5.458 \cdot 10^{-2}$	$3.590 \cdot 10^{-3}$	no

Table 1: Observable replacement battery and admissibility-gate result.

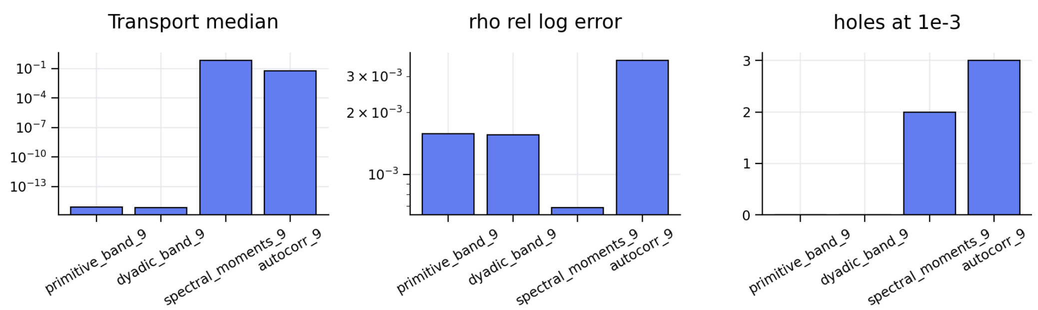

Figure 1: Admissibility-gate metrics for replacement observables.

5 Interpretation

The strong coordinate-level uniqueness theorem is false: there is not a single compelled coordinate list. The supported theorem is class-level uniqueness: admissible spectral representations define the same kind of transport-stable placement geometry, while non-admissible summaries fail either transport, null, or placement criteria.

6 Final Definition for Future Papers

Future papers should define W_{obs} as $[W_{\text{obs}}]_{\text{adm}}$, not as one handcrafted vector. A concrete coordinate basis may be used for computation, but claims attach to the admissible class.

7 Reproducibility

The outputs are `rg_admissible_transport_uniqueness_results.json`, `rg_admissible_transport_uniqueness_summary.txt`, and `run_rg_admissible_transport_uniqueness.py`. They build on `rg_observable_replacement_battery_results.json`.